

NinaivX — Literature Survey, in Plain English

A simple summary of all 29 works — what each is and why it matters to my project · Ashwin Kumar Ramesh Kumar · MSc AI, University of Sheffield

This one-look table explains, in everyday language, each source cited in the Literature Survey: what it is, and why it matters to NinaivX. Reference numbers [n] match the survey and its reference list.

Ref	Work (author, year)	What it is — in simple words	Why it matters to NinaivX
A · Foundations of conversational AI			
1	ELIZA (Weizenbaum, 1966)	The first-ever chatbot — faked a therapist using simple word-matching rules, with no real understanding.	Shows people bond with machines that don't truly understand them (the "ELIZA effect") — a real risk in a grief app.
2	Transformer (Vaswani et al., 2017)	The AI design ("attention") that powers all modern language models.	The foundation that makes today's realistic conversational AI possible.
3	GPT-3 (Brown et al., 2020)	A very large model that writes fluent, human-like text from just a few examples.	Powerful, but can confidently make things up — dangerous when imitating a real person.
4	BERT (Devlin et al., 2019)	An AI good at understanding text (not generating it).	Useful for understanding tasks, but can't produce the spoken replies my app needs.
B · Voice technology (speech in & out)			
5	Tacotron 2 (Shen et al., 2018)	Turns text into near-human-sounding speech.	A milestone in realistic text-to-speech — but only one fixed voice, no cloning.
6	Whisper (Radford et al., 2023)	Converts speech into text across many accents and noisy conditions.	The kind of speech-to-text that lets users simply talk to the app.
7	VALL-E (Wang et al., 2023)	Clones a person's voice from as little as 3 seconds of audio.	Proves voice cloning is possible — and raises serious consent concerns.
8	ElevenLabs (company, 2022)	A commercial service for AI voice generation and cloning.	The voice engine my app actually uses for speech and cloning.
C · AI companions & the loneliness problem			
9	WHO report (2025)	Says about 1 in 6 people worldwide are lonely, linked to ~871,000 deaths a year.	Proves the loneliness problem my companion mode addresses is real and huge.
10	Cigna Loneliness Index (2018)	Found ~46% of Americans sometimes or always feel alone.	More evidence that loneliness is widespread.
11	Xiaolce (Zhou et al., 2020)	A hugely popular empathetic chatbot, engineered to keep people chatting for a long time.	Shows companions work — but an engagement-first design can be unhealthy.

12	Ta et al. (2020)	Study: people report real companionship and support from the app Replika.	Evidence that AI companions can genuinely reduce loneliness.
13	Skjuve et al. (2021)	Interviews showing people form real relationships with chatbots.	Emotional bonds do form — though from a small, self-selected study.
14	Replika (company, 2017)	A popular AI companion app — which actually began from a chatbot built from a dead friend's texts.	A direct competitor, and an early link between companionship and grief.
15	Laestadius et al. (2024)	Study finding people become emotionally dependent on Replika, sometimes with harm.	Warns companion apps can create dependency — my design deliberately avoids this.
D · Digital afterlife & griefbots (recreating a person)			
16	HereAfter AI (company, ~2019)	Records a living person's stories, then replays them later in their own voice.	Competitor: safe but only replays recordings — it can't hold a real conversation.
17	Project December (Fagone, 2021)	News story of a man who recreated his late fiancée as a GPT-3 chatbot.	Famous case showing both the deep need and the danger (it can fabricate memories).
18	Microsoft patent (2020)	A patent for building a chatbot of a specific person — explicitly including the deceased.	Shows big firms see value here — but it's just a patent, unbuilt, with consent unresolved.
26	You, Only Virtual (company)	A commercial app to phone an AI version of a dead loved one; its founder wants to "solve" grief.	My closest competitor — I take the opposite stance: support healthy grieving, not eliminate it.
27	BBC "AI Confidential" (2026)	Documentary featuring YOY; Hannah Fry hears her own cloned voice, is moved to tears, yet calls it "a lie."	Real-world evidence of the emotional power — and the ethical debate my project engages.
E · Making it safe (stopping made-up answers)			
19	Hallucination survey (Ji et al., 2023)	A survey of AI "hallucination" — when a model makes up false information.	Names the core safety risk that my air-gapped design is built to reduce.
20	RAG (Lewis et al., 2020)	A method that grounds answers in real documents to reduce made-up facts.	The technique behind "grounding" the deceased persona in stored data.
21	RAG survey (Gao et al., 2023)	A survey organising the many ways to do retrieval-grounded generation.	Overview of deeper grounding methods — a clear avenue for my future work.
22	Llama Guard (Inan et al., 2023)	A safety filter that screens what goes into and out of an AI.	A guardrail approach — but generic, not tailored to grief.
28	LM harms taxonomy (Weidinger et al., 2021)	A structured map of the many harms language models can cause.	Frames hallucination as one of several harms a persona-imitating system must handle.
F · Ethics of recreating the dead			
23	Öhman & Floridi (2017)	Critique of the "digital afterlife industry" that monetises data of the dead.	Ethical foundation for how a dead person's data should be handled.

24	Öhman & Floridi (2018)	Argues that digital remains deserve dignity and protection.	Supports my privacy-first, respectful design.
25	Hollanek & Nowaczyk-Basińska (2024)	Proposes safeguards for griefbots: transparency, consent, and dignified "retirement".	The most relevant ethics guidance — my AI-disclosure and consent choices follow it.
29	Ethics of "deathbots" (Lindemann, 2022)	Argues recreating a person creates duties both to the dead and to the grieving.	Reinforces the ethical duties my design is built to respect.

Reference numbers match the Literature Survey and its reference list (all 29 verified against primary sources, June–July 2026).